Subprograms: Lambdas, Closures, Generators, and Coroutines

Programming Languages

Millersville University

Outline

- Lambda Functions
- Closures
- Re-entrant Subprograms
  - Generators
  - Coroutines

Lambda Functions

Lambda Functions

- **Lambda Function**, also called
  - function literal
  - lambda abstraction
  - anonymous function
  - lambda expression

- Originate from *Alonzo Church* and his invention of Lambda Calculus around 1936
  (all functions were anonymous)

$(\lambda x.x^2-2x+5)\ 2$

*In programming languages since **1958***

Lambda Function Tour

Lisp

```
(lambda () sexpr)
(lambda (x y) (+ x y))
```

JavaScript

```
() => expr
() => { stmts… }
(x, y) => x + y
```

Lambda Function Tour

OCaml

```
fun () -> ...
fun x y -> x + y
```

Java

```
() -> expr
() -> { stmts… }
(x, y) -> x + y
(x, y) -> { return x + y; }
(double x, double y) -> { return x + y; }
```

Lambda Function Tour

Python

```
lambda: expr
lambda x, y: x + y
```

Swift

```
{ () in expr }
{ () in stmts… }
{ x, y in x + y }
{ (x: int, y: int) -> int in return x + y }
{ $0 + $1 }
```

Lambda Function Tour

C++

```
[]() { stmts… }
[](auto x, auto y) { return x + y; }
[](int x, int y) -> int { return x + y; }
```

Ruby

```
lambda { expr }
lambda { stmts… }
lambda { |x, y| x + y }
-> (x, y) { x + y }
```

Closures

Closures

- A *closure* is a subprogram AND *referencing environment* for where it was defined
  - Key difference from lambdas: can access *non-local vars*
- Most often written as a lambda with no syntactic difference
  - Lisp
  - Java
  - JavaScript
  - OCaml
  - Python
  - Swift
  - Ruby

```
var base = 10
let f = { (x: Int) -> Int in
    return x + base
}
```

- Exception: C++

Closure Example

JavaScript

```
x => y => x + y
x => { return y => x + y }
                       ^
```

OCaml

```
let multiply n lst =
  List.map (fun x -> n * x) lst
                         ^
```

Closure Example

C++

```
namespace r = std::ranges;
std::vector<int>
multiply (int n, std::vector<int> list) {
    r::for_each (list, [n] (int& value) {
    return value * n; //^ explicit copy of n
  });
  return list;
}
```

Closure Example

C++

```
namespace r = std::ranges;
std::vector<int>
multiply (int n, std::vector<int> list) {
    r::for_each (list, [=] (int& value) {
    return value * n; //^ all non-locals are
  });                 //  passed by value
  return list;
}
```

Closure Example

C++

```
namespace r = std::ranges;
std::vector<int>
multiply (int n, std::vector<int> list) {
    r::for_each (list, [&] (int& value) {
    return value * n; //^ all non-locals are
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}
```

Coroutines

Subprogram Definition

A subprogram has…

- Exactly one entry point
- One or more exit points

Subprogram Definition

A subprogram has…

- Exactly one entry point
  - The beginning of the function!

- One or more exit points
  - Can have return statements throughout
  - Exceptional control flow (later in course)

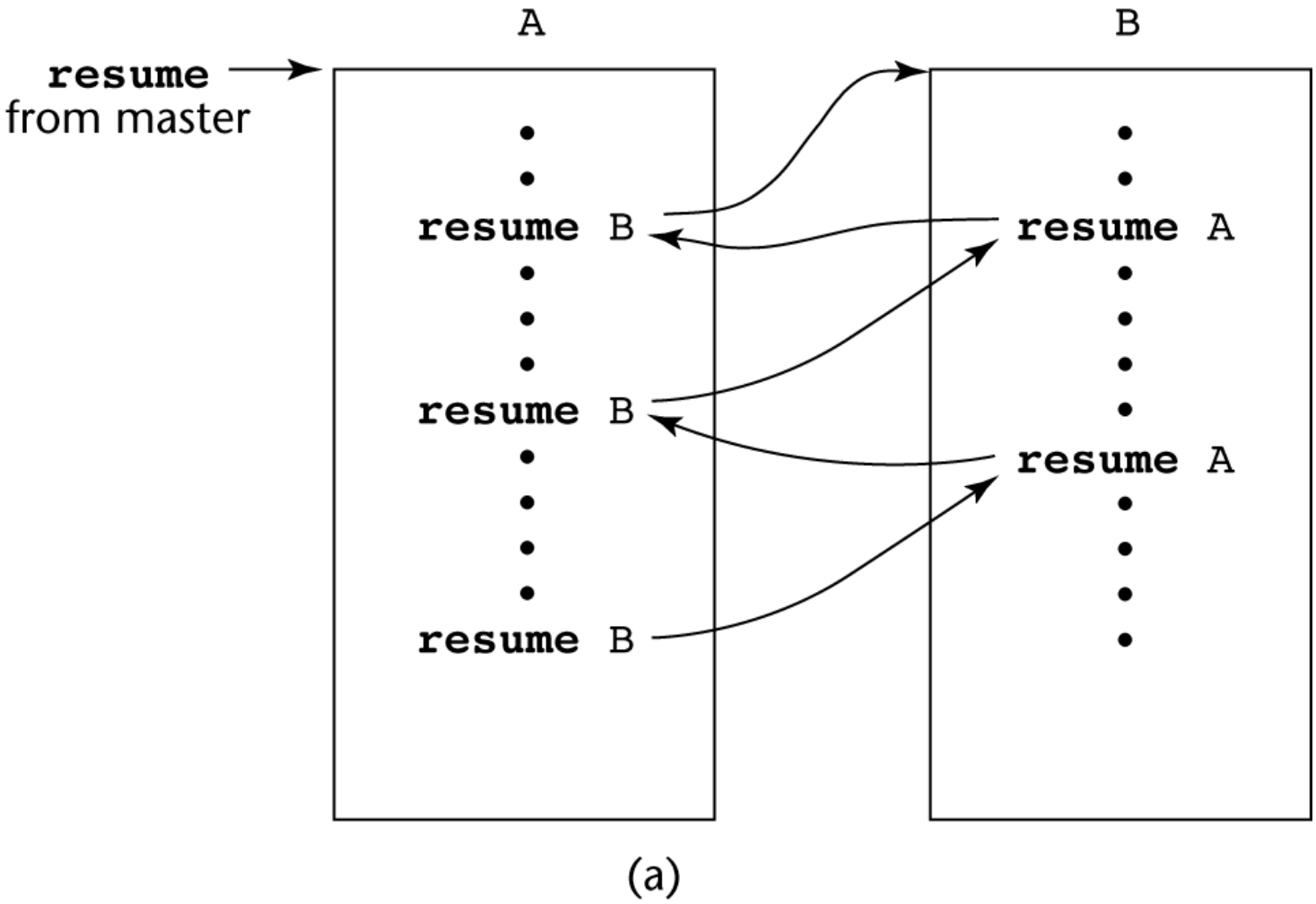
Coroutines

- A *coroutine* is a subprogram that has multiple entry points
- Also called *symmetric control*
  Caller and callee coroutines are on a more equal basis
- A coroutine call is named a *resume*
  - First resume of a coroutine is to its beginning
  - Subsequent calls pickup where left off
- Coroutines provide *quasi-concurrent execution*
  - Good for concurrent programming; multiple tasks make progress, but only one runs at a time
  - Execution is interleaved, but not overlapped

A coroutine is a subprogram that can:

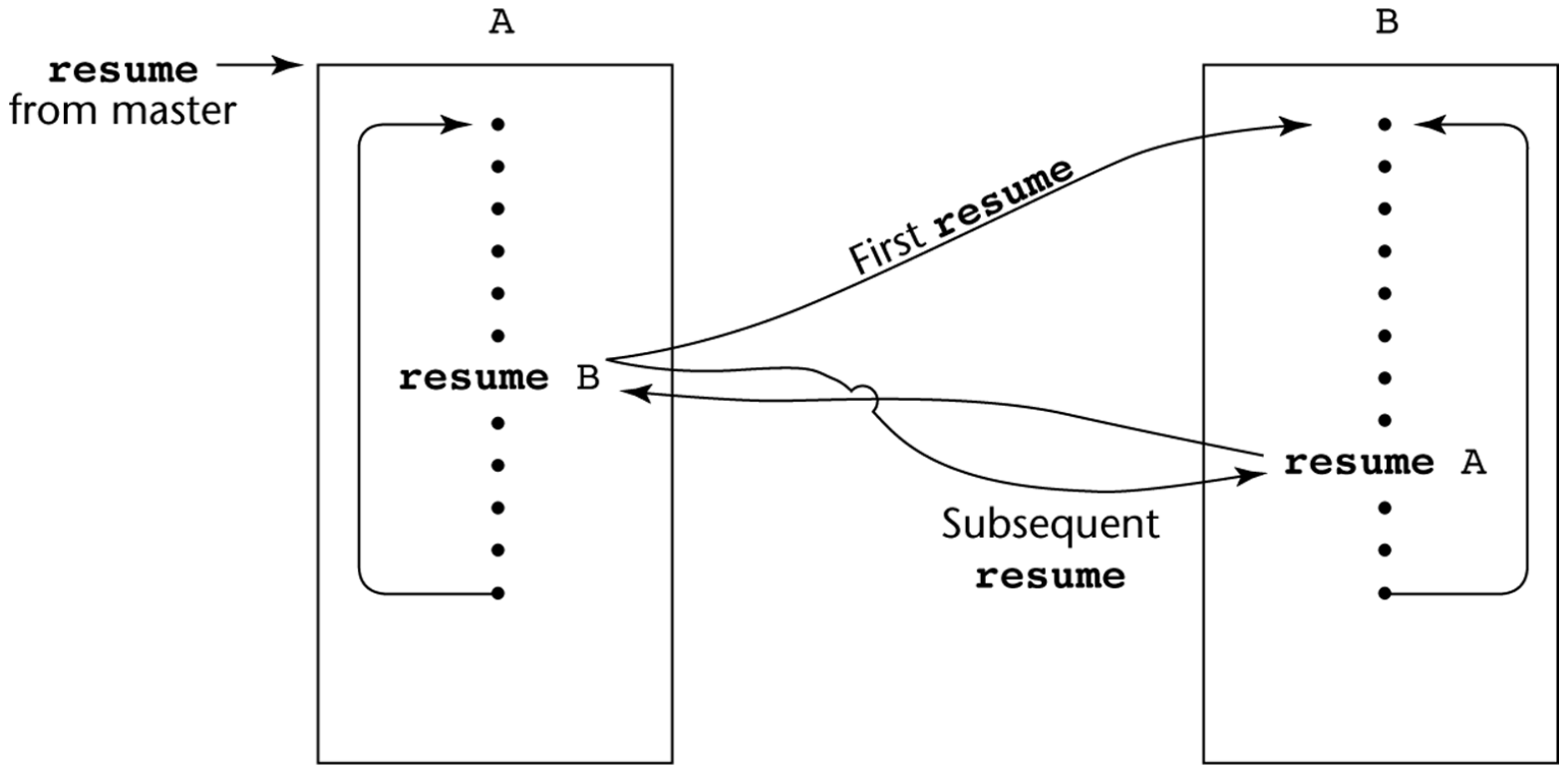
- pause execution
- save its state
- resume later from the same point

Coroutines Illustrated: Possible Execution Controls



(a)

Coroutines Illustrated: Possible Execution Controls with Loops



Coroutines: Language Support

- C++ (since C++20)
- C#
- D
- F#
- Go
- JavaScript
- Julia
- Lua
- PHP
- Prolog
- Python
- Ruby
- Rust
- Scheme (Lisp-like)

Coroutines Example

```
def match(pattern):
    print('Looking for ' + pattern)
    try:
        while True:
            s = (yield)
            if pattern in s:
                print(s)
    except GeneratorExit:
        print('Done.')
```

Coroutines Example

```
>>> matcher = match('hello')
>>> next(matcher)
    Looking for hello
>>> matcher.send('hello there')
    hello there
>>> matcher.send('goodbye now')
>>> matcher.send('Othello is a great play')
    Othello is a great play
>>> matcher.close()
    Done.
```

```
def match(pattern):
    print('Looking for ' + pattern)
    try:
        while True:
            s = (yield)
            if pattern in s:
                print(s)
    except GeneratorExit:
        print('Done.')
```

yield → pauses execution
.send(value) → resumes and sends data to s

Generators

- Generators are an example of coroutines
- Generators can give us different values each invocation time
- Values are not computed when the generator is created, but when they are requested!

Generators

Goal

- Don’t want to exit the subprogram – simply “pause” it in some way

- Achieved by **not** using `return`

- Define a new control flow keyword!

Generators

- `yield` values

- Yield “pauses” execution of the subprogram

- When we invoke subprogram we resume from where we left off

Python:

```
def first_three():
    yield 1
    yield 2
    yield 3
```

Generators

```
def ones():
    while True:
        yield 1

def natural():
    x = 0
    while True:
        yield x
        x += 1
```

```
gen = natural()
next(gen)
next(gen)
next(gen)
next(gen)
```

Both next() and .send() resume the coroutine.

Generators

```
# Exercise
def fibonacci():
    a, b = 0, 1
    while True:
        ......
```

Recursive Generators

```
from os import listdir
from os.path import isfile, join

def print_files(dir):
  for file in listdir(dir):
    full_path = join(dir, file)
    if isfile(full_path):
        yield full_path
    else:
        yield from print_files(full_path)
```

Attribution

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Sources include Robert Sebesta's Concepts of Programming Languages